Machine Learning and pattern recognition

Fingerprint spoofing detection 2024

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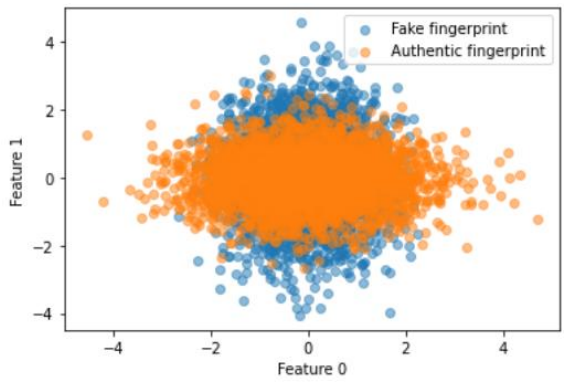
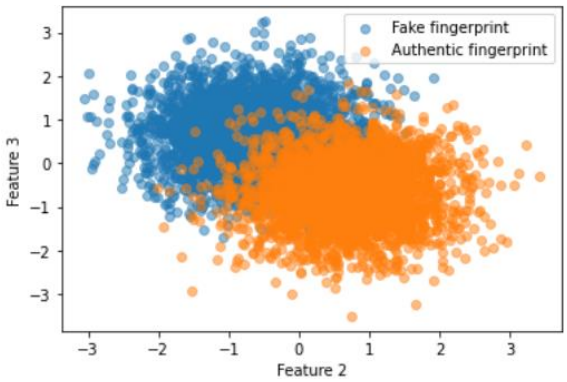
INTRODUCTION

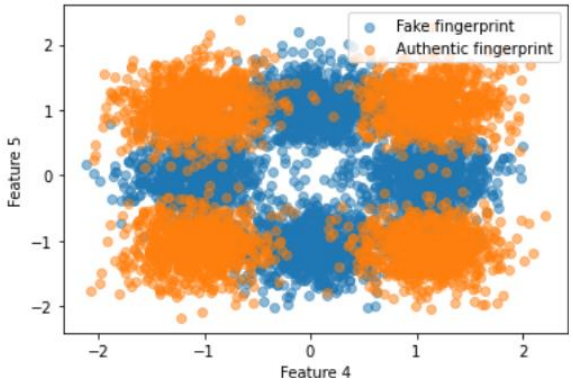
This report has the goal to examine a dataset that contains various images of Authentic and Fake fingerprint, it is quite balanced and composed by 6000 samples, 3010 belonging to the authentic class and labelled as 1 and 2990 belonging to the fake one and labelled as 0. Each sample is represented by 6 features, continuous values that represent a point in a multi-dimensional space, which summarizes high-level characteristics of a fingerprint image. We start by first exploring and analysing the structure of the dataset trying to understand how it is distributed. Then, we will evaluate various classifiers and try to find the best system that can accurately classify our samples.

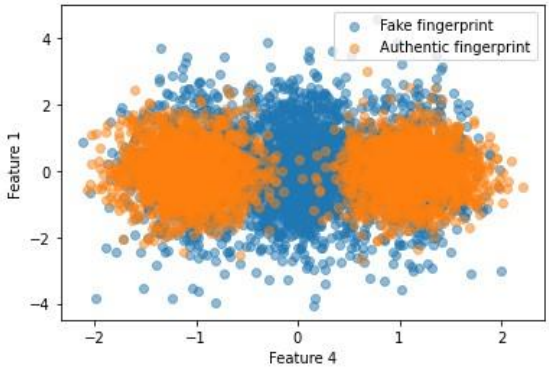
Capitolo 1:
FEATURE ANALYSIS

We start analysing the features of our data. Here there is a series of plot about the features distribution.

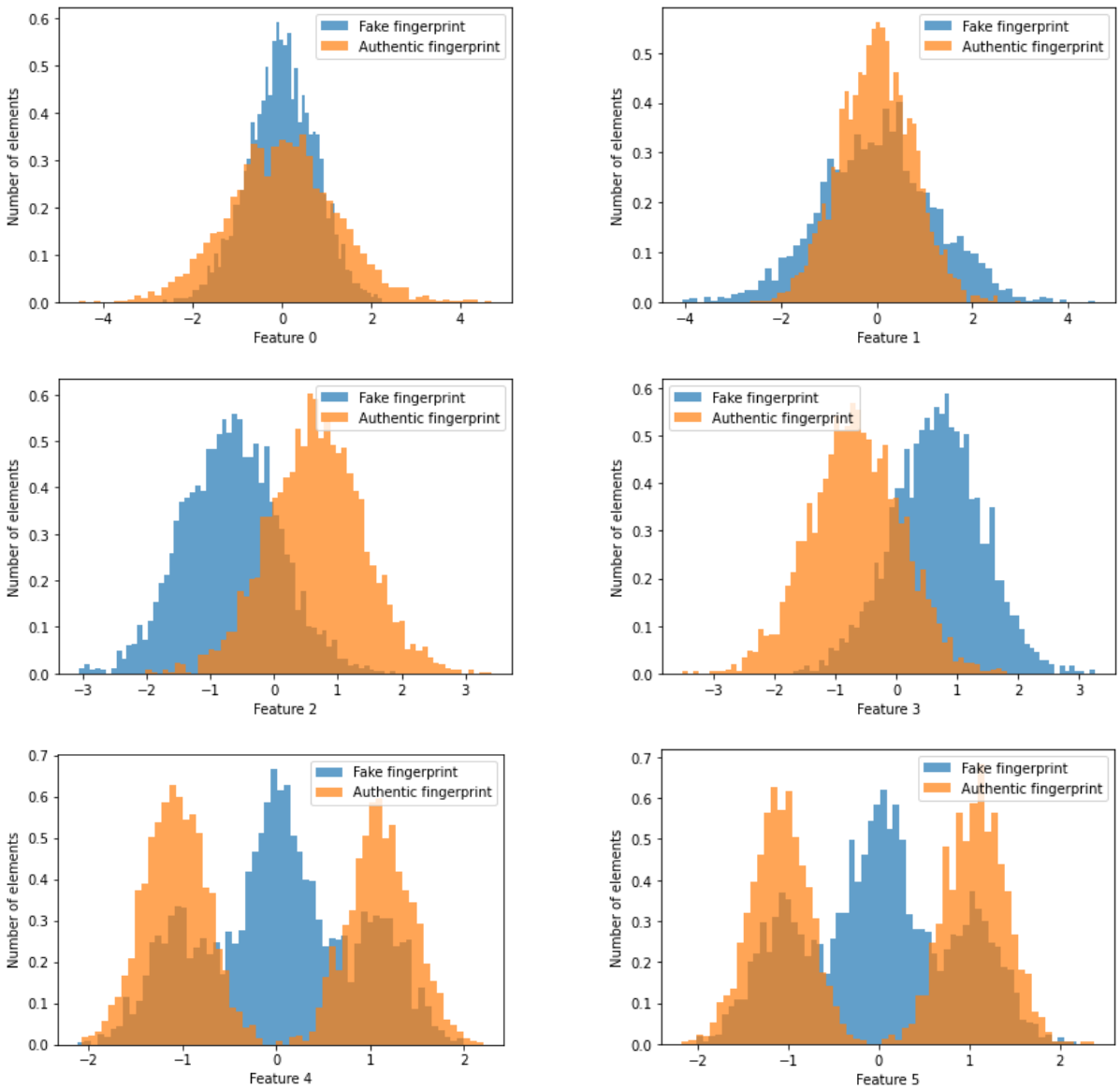








We can notice from the scatter plots of some of the most relevant features that for the two class we have a different number of clusters for each class that goes from one to four. We can also see that there is a large overlap that cannot help us to well separate the samples of the two class and suggest us to use techniques of dimensionality reduction to better classify the samples.

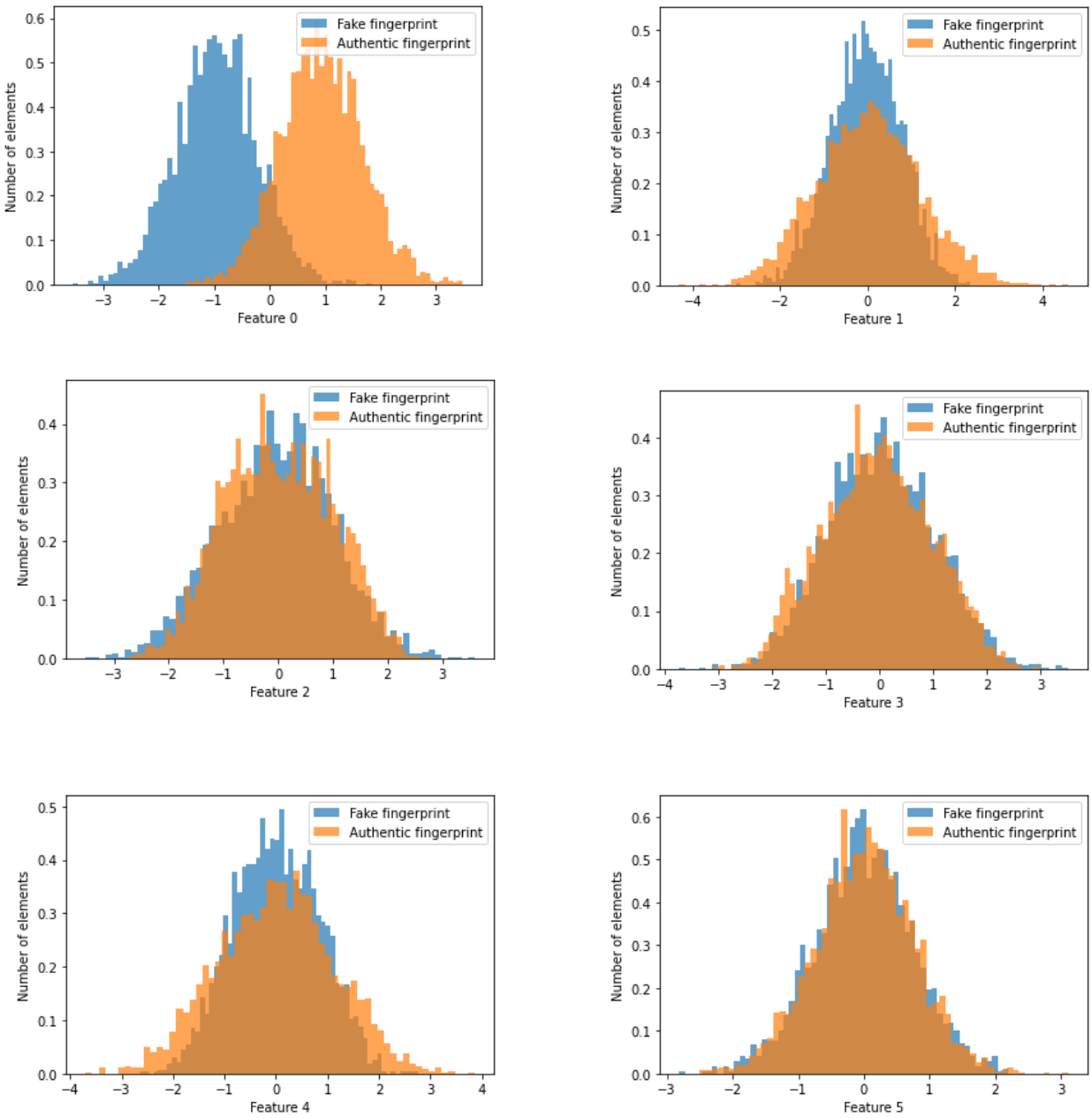


These are the histograms of the different features. Let's consider features 0 and 1. These features exhibit mean values close to zero for both the Authentic and Fake classes. However, we observe, for the second feature, a notable difference in variance between the two classes. The variance is lower for the Authentic class, indicating that the values of these features are more tightly clustered around the mean compared to the Fake class. Moving on to features 2 and 3, we encounter an interesting scenario. The mean values of these features show similar means but with opposite signs between the two classes. This implies that these features might play a significant role in discriminating between Authentic and Fake fingerprints. Moreover, the similarity in variance suggests a comparable spread of data around the respective means for both classes. Now, let's discuss the last two. These features exhibit similar mean and variance values across both classes. However, they do not demonstrate as

clear a distinction between Authentic and Fake fingerprints compared to features 2 and 3.

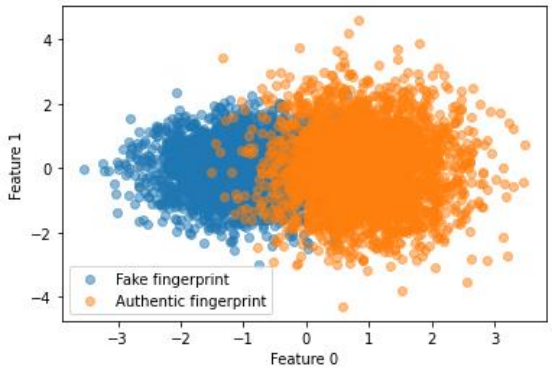
1.1 **Dimensionality Reduction**

We now turn our attention to applications of techniques like PCA and LDA to our dataset. Both are linear transformation techniques that are commonly used for dimensionality reduction. We start our analysis applying PCA. Here we plot the histogram of the projected features over the six directions:



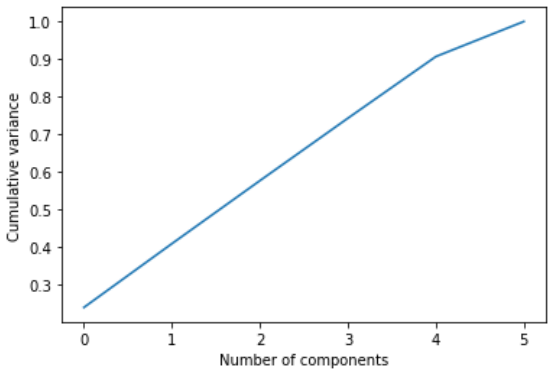
We can observe that only for the first direction, the one with the most variance, there is less overlap, and this suggest us that this component could represent some relevant characteristic for our binary

classification while for the other principal component is quite difficult to distinguish between the fake and the authentic fingerprint. This impression is confirmed by the scatter plot below:



PCA

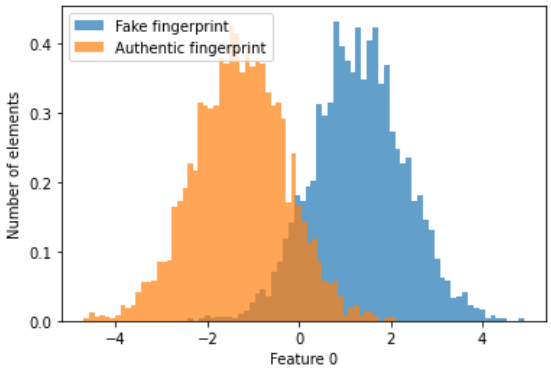
We focus now on the correlation between the features using Pearson correlations coefficients. We do this analysis to better visualize the relationship between the different features for both classes. What we can deduce is that the almost all features are poorly correlated. Only the feature 4 and 5 for the Fake and Authentic class are quite correlated and this suggests that a reduction in the dimensional space from 6-dimensions to 4-dimensions could bring benefits.



Cumulative variance over PCA dimensions

From the graph above we see that with PCA considering the first four principal components we store up to 0.9% of information.

Now we turn attention on applying LDA to the dataset. Looking at the graph we see how it is possible to use linear separation methods for the two classes, but the separation would not be without errors.

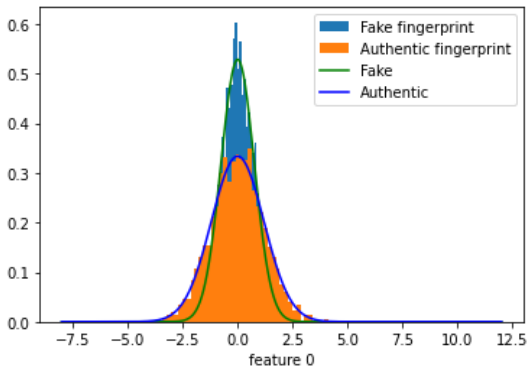


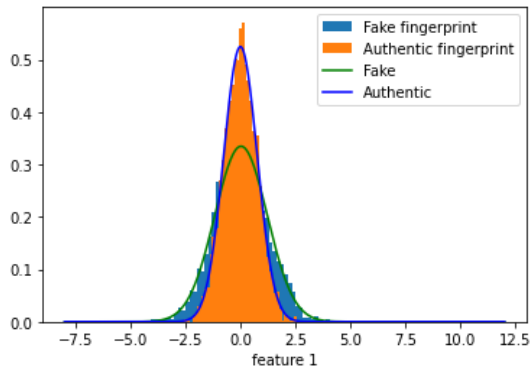
LDA

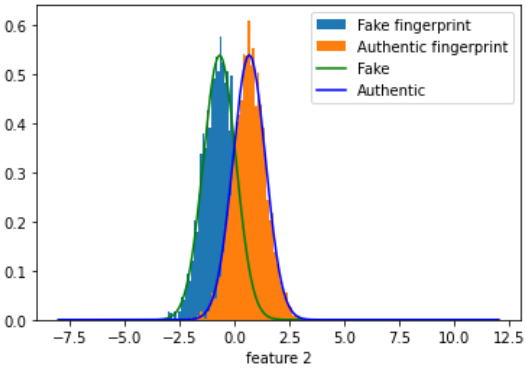
Initially LDA was popular also as classifier before other more complex techniques as non-linear machine learning algorithm were introduced. We divide the dataset in training set (DTR-LTR) and validation set (DVAL-LVAL) and apply LDA so that it selects the orientation that results in the projected mean of class True (label 1) being larger than the projected mean of class False (label 0). By first calculating a threshold as the mean of the projected means is possible to compute the predictions on the validations data and the accuracy of the LDA classifier. In our case the accuracy is of 90.7 % that is a good result. By choosing other values for the threshold the results are the same. It's common also to use both PCA and LDA in combination. In our project we apply first PCA on the train data and then classify the validation data by applying LDA. Trying with different value for the number of principal component (PCA) is clear that in this task the combination of both the techniques doesn’t provide benefits because the accuracy, with the m = 2 (best value for the number of components) give us only a 0.05 % rate of improvement in the classifier accuracy.

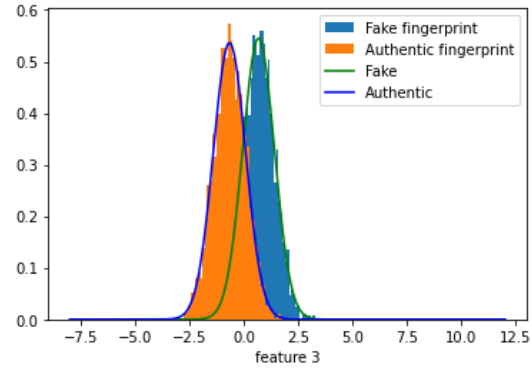
1.2 **Uni-variate Gaussian Density**

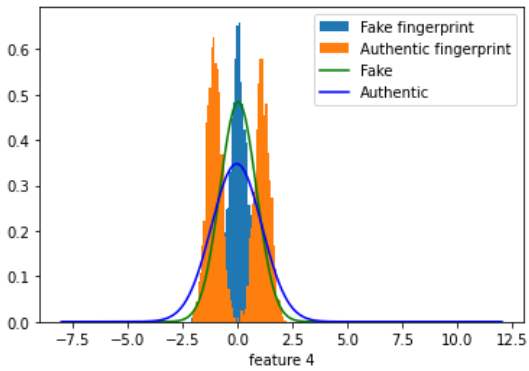
We now try to inspect if there are features for which the Gaussian density provide a good fit. The analysis has been done for the different features of each class and below we have the plots:

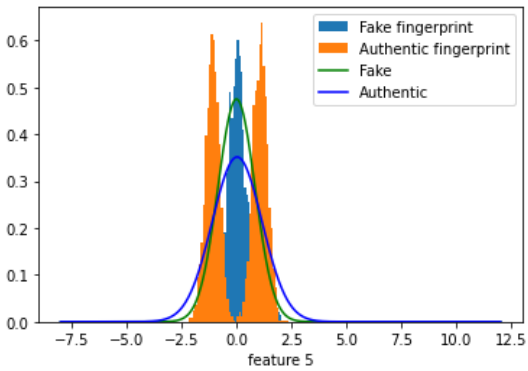












It ‘evident that for the first four (feature 0 to feature 3) both the class can be approximated with a Gaussian distribution especially the Authentic one , while for the last two the data are quite spread and the Gaussian distribution seems significantly less accurate and doesn’t provide a good fit.

Capitolo 2:
CLASSIFICATION AND VALIDATION

2.1 GAUSSIAN MODELS

We are now ready to begin our examination of machine learning models by looking at Gaussian classifiers such as the full covariance (MVG), the diagonal covariance Naive Bayes, which assumes that assumes that covariance matrices Σ are diagonal matrices, and the Tied model, which assumes that the covariance matrices are the same for all classes.

We start by training the parameters (μ, Σ) on the training portion of the dataset and then we compute the loglikelihood ratio (LLR) on the validation set. The analysis is done considering first the entire set of features and then only the first four features. This are the result for the three models:

MODEL	ERROR RATE (ALL 6 FEATURE)	ERROR RATE (FEATURE 0-3)
MVG	7.00%	7.95%
NAIVE BAYES	7.19%	7.64%
TIED MVG	9.30%	9.50%

We obtain that the MVG and the NAIVE BAYES performs better, and the worse performance is obtained from the TIED MVG. This is reasonable since we have seen that the first four feature of both classes can be approximated with a Gaussian distribution and are poorly correlated. However, the results are not so bad for the tied one and it's reasonable since we have seen from the LDA plot that a linear separation is also good for our dataset. What is evident is that discarding the last two features keeps the error rate quite similar for the three models and this is something that we could expect from the plot of the Gaussian distribution we did in the last chapter.

We now apply the MVG and TIED MVG models using only feature (0-1) and feature (2-3):

MODEL	ERROR RATE (FEATURE 0-1)	ERROR RATE (FEATURE 2-3)
MVG	36.50%	9.44%
TIED MVG	49.00%	9.40%

For the first case (feature 0-1) the MVG model continues to be the better one although the error rate increased a lot, and this is due to both to the overlap we observed when we did the plot of the histograms and the scatter plots of the two features as well as to the values of variance that showed really different values, in contrast specially with the hypothesis of class-independent noise of the TVG model . For the second case (feature 2-3) the two models show similar error rate. The MVG is worser than the case in which we used only the first four feature, while the TIED MVG shows a little improvement in the classification, due to the similar variance showed by these two features.

Last analysis is done by first pre-processing the dataset with PCA and then applying the three different classifiers:

MODEL	PCA	ERROR RATE
MVG	2	8.79%
	3	8.79%
	4	8.05%
	5	7.10%
NAÏVE BAYES	2	8.85%
	3	9.00%
	4	8.85%
	5	8.75%
	6	8.90%
TIED MVG	2	9.25%
	3	9.25%
	4	9.25%
	5	9.30%

As expected, the better performance is obtained by PCA but also keeping 5 dimensions for all the three classifiers doesn't bring benefits since the performance are worse. For Naive Bayes also results for PCA equal to 6 are reported because performing PCA and keeping the same number of dimensions as the one of the original features diagonalizes the covariance matrix of the dataset, but in this case is more accurate the model without PCA. For the others results with PCA equal to 6 were the same obtained without PCA so we omitted them.

2.2 **Validation methodology**

Up to now we focused only on assigning labels to the samples based only on maximum-a-posteriori class probabilities, without considering that different error could have different impact. Our goal is always that of evaluate the performance of our classifier and for that reason we have used some metrics ad the accuracy or, equivalently, the error rate. However, they can be misleading if the classes are not balanced, so to overcome this issue we employ an evaluation set to compute the performance of different models.

We start to analyse the performance of the Gaussian classifier considering different applications given by this triplet $(\pi_1; Cfn, Cfp)$, that represents the working point of an application for a binary classification task. However, the working point is overparametrized, in fact we can show that s possible to build infinite equivalent applications which have the same decision rule as the original one, but different costs and priors.

We so focused on one application $(\tilde{\pi}; 1,1)$, where we can interpret $\tilde{\pi}$ as an effective prior, that reflects the importance of cost of misclassification in the decisional process of the classifier, and the costs are uniform. A lower effective prior $(\tilde{\pi} = 0.1)$ occurs when the cost of a false positive (Cfp) is higher than the cost of a false negative (Cfn). This mean that the classifier will be more conservative in labelling a sample as genuine, because the acceptance of an impostor is more expensive. The opposite verifies when we have a higher effective prior $(\widetilde{\pi} = 0.1)$ that leads the classifier to be more permissive, labelling more easily sample as genuine.

2.3 **Gaussian classifier performance**

We now focus on three applications, represented in terms of effective priors:

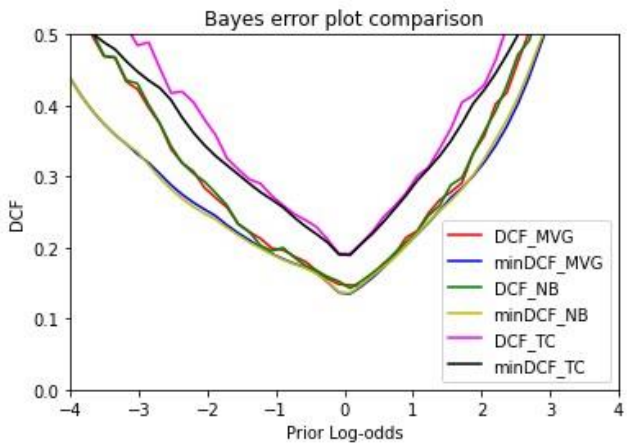
MODEL	PCA	ACT. DCF $(\tilde{\pi} = 0.1)$	MIN. DCF $(\tilde{\pi} = 0.1)$	ACT. DCF $(\tilde{\pi} = 0.5)$	MIN. DCF $(\tilde{\pi} = 0.5)$	ACT. DCF $(\tilde{\pi} = 0.9)$	MIN. DCF $(\tilde{\pi} = 0.9)$
MVG	None	0.305	0.263	0.140	**0.130**	0.400	**0.342**
	5	0.304	0.274	0.142	0.133	0.398	0.351
	4	0.353	0.301	0.161	0.154	0.460	0.415
	3	0.388	0.356	0.176	0.173	0.468	0.439
	2	0.388	0.353	0.176	0.173	0.443	0.438
	1	0.397	0.369	0.185	0.177	0.478	0.434
NAIVE BAYES	None	0.302	**0.257**	0.144	0.131	0.389	0.351
	6	0.392	0.353	0.178	0.173	0.451	0.436
	5	0.393	0.354	0.175	0.174	0.466	0.434
	4	0.397	0.361	0.177	0.172	0.463	0.431
	3	0.395	0.365	0.180	0.175	0.459	0.434
	2	0.387	0.356	0.177	0.171	0.442	0.432
	1	0.397	0.369	0.185	0.177	0.478	0.434
TIED MVG	None	0.406	0.363	0.186	0.181	0.463	0.442
	5	0.405	0.365	0.186	0.181	0.463	0.445
	4	0.403	0.361	0.185	0.182	0.462	0.444
	3	0.408	0.368	0.185	0.183	0.457	0.434
	2	0.396	0.363	0.185	0.179	0.479	0.435
	1	0.402	0.369	0.187	0.177	0.481	0.434

As expected, the MVG model perform better than the other two models with a minDCF that is the lowest for all the applications except for $\tilde{\pi} = 0.1$ where the Naive Bayes behave a bit better. The Tied MVG is that with the worse performance and it is something that we already knew from the previous analysis. In terms of calibration both MVG and Naïve Bayes are well calibrated, with a difference

between minDCF and actDCF in the order of few percentage points. The Tied Covariance instead, show a higher calibration loss, but almost acceptable.

We will now consider from now on as target application the one with $\tilde{\pi} = 0.1$: for the MVG and the Naive Bayes the lowest minDCF is obtained without using PCA as pre-processing, while for the Tied MVG we have a minor cost when applying PCA with m equal to 4.

Considering the configuration that gave us the best result for each model we use the Bayes Error Plot to assess their performance as we vary the application:



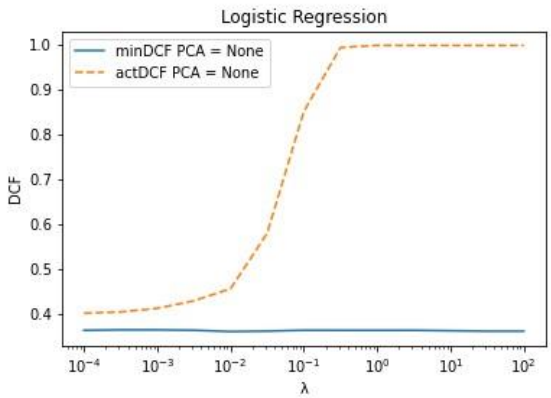
From the graph above we can observe the for low prior log-odds the DCF and minDCF for TC are relatively higher, indicating poor performance compared to MVG and NB; for prior log-odds in the range -1 to 1 the DCF and minDCF for MVG are the lowest; for hight prior log-odds NB tend to perform well but still MVG has slightly better performance.

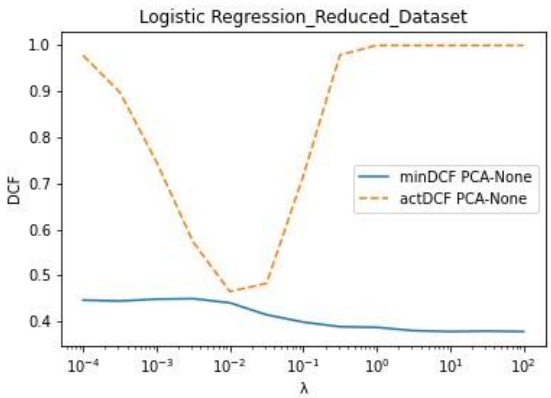
Considering all the range MVG and NB show good calibrations since DCF and minDCF curves are close to each other's across most of the prior log-odds range, while the TC show poor calibration with a noticeable gap between the curves.

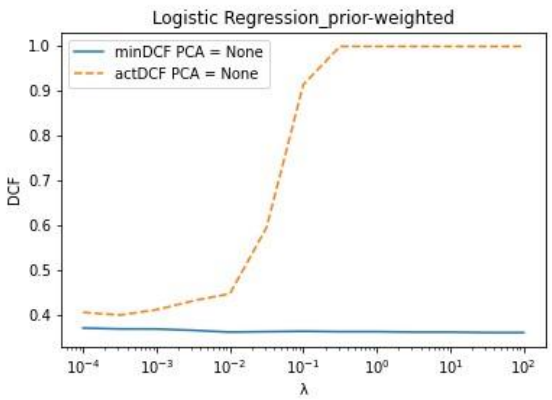
2.4 **Logistic Regression**

We shift now our attention first to Logistic Regression with is linear separation rule of data and we apply both the standard version of the model, which reflects the empirical class prior of the training data, and the prior-weighted version to simulate different target prior, in particular we analysed the case of $\pi_t = 0.1$. We started by plotting minDCF and actDCF graphs for a range of values of λ, the regularization coefficient, to inspect which is a optimal value for it. We both applied the model to full dataset and to a reduced version of the dataset to better understand the role of regularization.

Below are reported the graphs:







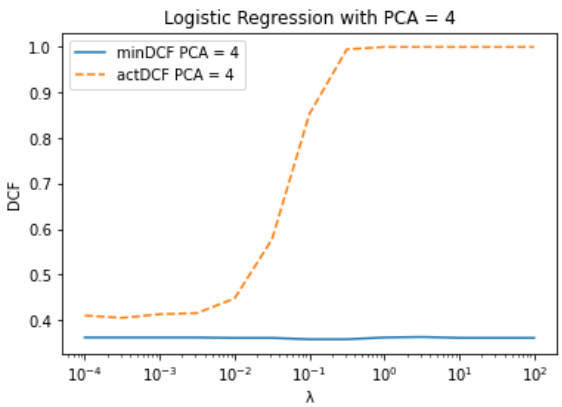
In the first graph (Logistic Regression), we observe that min DCF remains quite stable across all values of λ indicating that the performance of the model in terms of minDCF is not affected by regularization, while actDCF increases as λ increases particularly after $\lambda \approx 10^{-1}$, then stabilizes indicating that actDCF is sensitive to over-regularization.

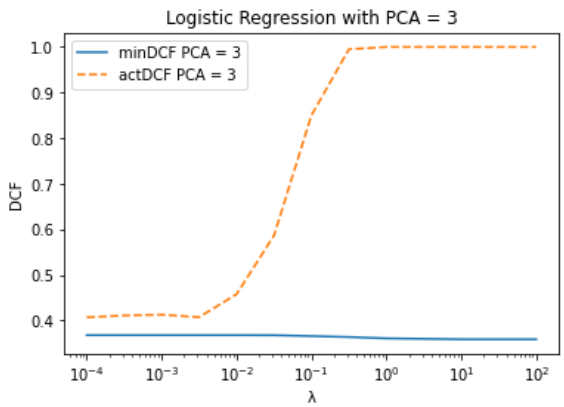
In the second graph (Logistic Regression with reduced Dataset) we can observe that minDCF decreases for values of $\lambda > 10^{-2}$ while actDCF reachs a minimum in $\lambda \approx 10^{-2}$ and then increases for larger values of λ with a degrades of performances. The last graph(Prior-Weighted Logistic Regression) is similar to the first we described indicating that in our case the prior-weighted version of the model doesn't give advantages.

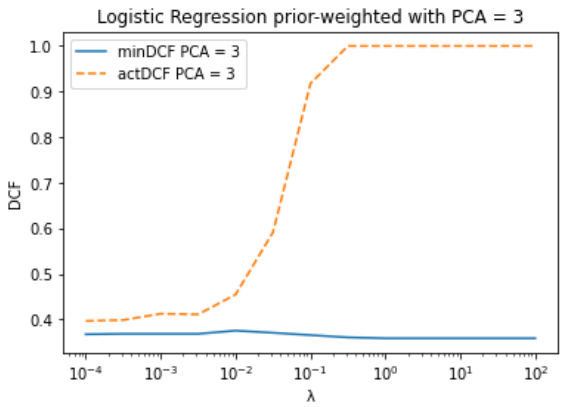
λ	MIN. DCF	ACT. DCF
10^{-4}	0.364	0.402
10^{-3}	0.365	0.413
10^{-2}	0.361	0.457
10^{-1}	0.364	0.852
1	0.364	1.0
10^{1}	0.363	1.0
10^{2}	0.362	1.0

Table 1: Logistic Regression

Since the regularized model is not invariant to the linear transformations of the feature vectors could be useful to apply some techniques as PCA on our dataset to see if this can lead to different results.





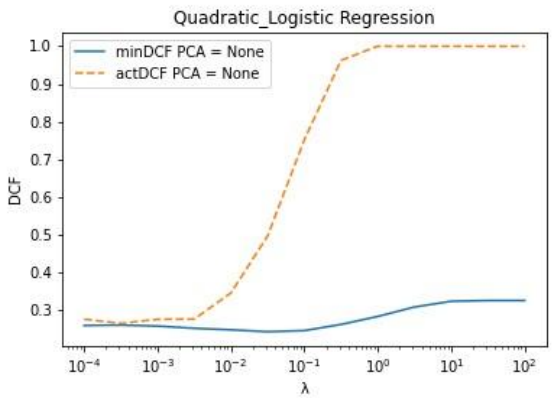


It's evident from the plots that PCA as preprocessing don't lead to great variations, in fact the curves show a similar behaviour to what we have seen before, and this is something that we could expect also from our previous analysis since PCA keeps all the necessary informations and variance relevant

for our model.

2.4.1 **Quadratic logistic regression**

We now focus on the Quadratic logistic regression. Given that the best results until now were achieved using the MVG model, we can expect to obtain good results also for the Quadratic logistic regression, as our data seem to be better separated by quadratic rules. What is done by this version of the model is to find an expanded feature space in which linear surfaces corresponds to quadratic surfaces in the original space.



From the graph we can observe that the lowest value of minDCF is obtained for $\lambda \approx 10^{-1}$ while when values of the regularization coefficient increase the curve growth leading to higher costs.

λ	MIN. DCF	ACT. DCF
10^{-4}	0.260	0.277
10^{-3}	0.259	0.277
10^{-2}	0.249	0.346
0,031	**0.244**	0.497
10^{-1}	0.247	0.752
1	0.284	1.0
10^{1}	0.324	1.0
10^{2}	0.326	1.0

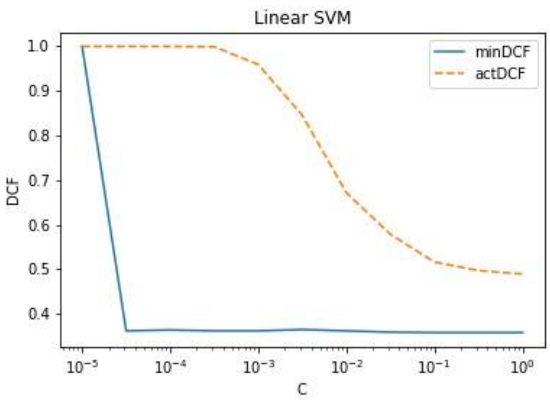
Table 2: Quadratic logistic regression

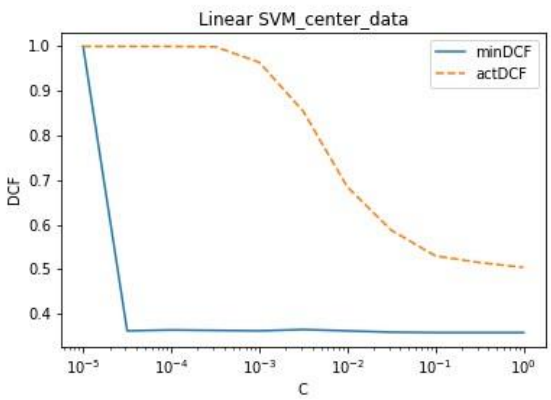
Finally observing the result, we can conclude that the results computed with the Quadratic logistic regression are better than the linear version. The best configuration is obtained with λ =0.031 better than any other considered model so far, obtaining relatively close but anyway lower minDCF that the one computed with the Gaussian classifiers.

2.5 Support Vector Machine

In this section we apply the Support Vector Machine model that allows us to extend the problem of the logistic loss seen with the Logistic Regression giving a geometrical interpretation of the regularization term. The goal of the SVM is to select the hyperplane that best separate the class with the largest margin, which means to maximize the minimum distance of all points from the hyperplane. The model that we applied to our dataset are the Linear SVM and two non-linear version: the RBF (Radial Basis Function) kernel and the Polynomial kernel.

2.5.1 Linear SVM





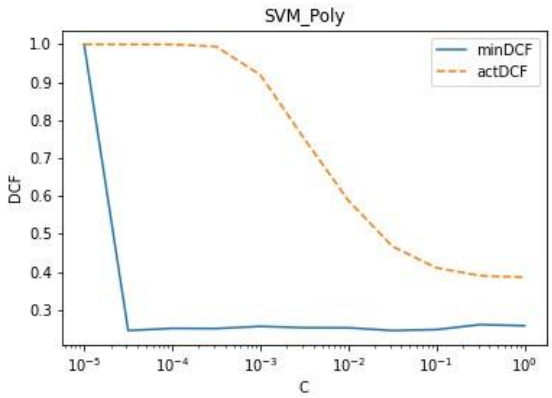
C	MIN. DCF	ACT. DCF
10^{-5}	1.0	1.0
10^{-4}	0.364	1.0
10^{-3}	0.362	0.959
10^{-2}	0.362	0.672
10^{-1}	0.358	0.516
1	0.358	0.489

Table 3: Linear SVM (K=1)

From the plot of the two curves of both graphs, we can observe that for very small values of C (strong regularization) the minDCF starts very high and then drop sharply. As C increases the minDCF remains almost constant. The actDCF starts also with high values for C small and then decreases but more gradually than the minDCF curve. Strong regularization leads to poor performance indicating underfitting while for weak regularization (large values of C) the results are better indicating more adaptability in fitting the data. What is evident is that there is a gap between the two curves suggesting that scores may not be well calibrated. Since SVM is not invariant to transformations of data we tried to apply the SVM also on centered data but in our case the behaviour doesn't change much. In conclusion we can say that for our application with $\tilde{\pi} = 0.1$ linear SVM are quite like other linear model we applied in terms of results. Something that we can expect is that the non-linear version of the SVM will perform better on our data.

2.5.2 **Polynomial kernel**

Let's turn our attention on Polynomial kernel to see if more complex separation surfaces are more capable to well separate our data. We consider only the kernel with d=2 and c=1.



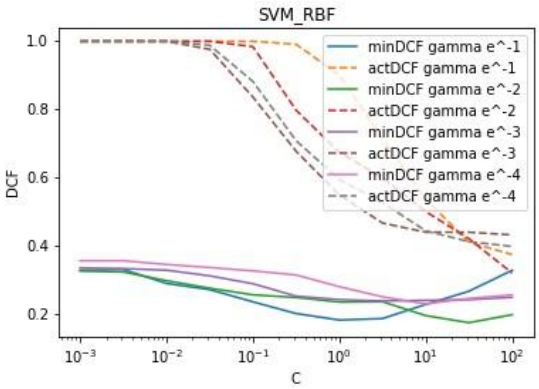
C	MIN. DCF	ACT. DCF
10^{-5}	1.0	1.0
10^{-4}	0.251	1.0
10^{-3}	0.257	0.92
10^{-2}	0.253	0.588
10^{-1}	0.248	0.411
1	0.258	0.386

Table 4: SVM Poly (K=0)

From the graph we can see that the two curves are very similar in their behaviour to the Linear SVM, but they have minor cost and especially the actDCF decrease rapidly. If we compare this model with the other non-linear model particularly the Quadratic Logistic regression, we can see that for the minDCF we obtain similar values and what is evident is that as for the values of λ in the Logistic Regression that for the values of C in SVM better result are obtained with values in the order of 10^{-1}. The actDCF instead is the inverse in the two-case analysed as the curve increase for increasing values of λ, while decrease for increasing values of C. As we have seen for the other model technique of pre-processing don't lead to better performance, so we don't report them.

2.5.3 **RBF Kernel**

We have already seen the polynomial kernels, but there exists also another important kernel which is associated to an infinite-dimensional mapping, the Gaussian Radial Basis Function Kernel (RBF). It depends in his formulation on different hyperparameters, i.e. C, K, γ, so we try to analyse the impact of different combinations of their values. Below are reported the graph and the best results we obtained (with $\gamma = e^{-2}$) :



C	MIN. DCF	ACT. DCF
10^{-3}	0.327	1.0
10^{-2}	0.298	1.0
10^{-1}	0.257	0.984
1	0.236	0.675
10^{1}	0.196	0.501
31,66	0.175	0.422
10^{2}	0.199	0.321

Table 4: SVM RBF (K=1, $\gamma = e^{-2}$)

As we expected applying more complex separation rule leads to lower cost, in fact for values of $\gamma = e^{-2}$ and higher values of C we have that both actDCF and minDCF tendo to be lower, indicating better performance. The scores are poor calibrated for lower values of C while the calibration improves for higher C. For our application this model reaches the best results compared to previous one and this is significantly important because suggest that RBF kernel is effective at capturing non-linear relationships in our data, however high gamma values, that indicate more complex model, can be synonymous of overfitting if not balanced with appropriate C values.

2.6 Gaussian Mixture Model

The last classifier that we will apply is the Gaussian mixture model, which assumes that we can represent data with Gaussian distribution composed of multiple components(clusters). The model that will be considered are the Full Covariance, the Diagonal Covariance and the Tied Covariance.
What we expect from the previous analysis of the MVG and Naïve Bayes model is that the assumption of uncorrelation of the feature lead to better performance for our data.
We tested the GMM models considering different number of components (up to 32) for the Authentic and the Fake class for all the three type of GMM (Full, Tied and Diagonal) and below are reported the some of the best results in terms of minDCF and actDCF :

FAKE	AUTHENTIC	MIN. DCF	ACT. DCF
8	2	0.181	0.191
16	1	0.167	0.178
16	2	0.166	0.175
8	16	0.153	0.172
1	16	0.15	0.206

Table 5: FULL COVARIANCE

FAKE	AUTHENTIC	MIN. DCF	ACT. DCF
4	1	0.145	0.15
4	8	0.140	0.152
4	16	0.137	0.169
8	16	0.132	0.149
8	32	0.131	0.152

Table 6: DIAGONAL COVARIANCE

FAKE	AUTHENTIC	MIN. DCF	ACT. DCF
2	8	0.155	0.187
2	16	0.159	0.197
8	16	0.162	0.181
16	16	0.164	0.190
4	32	0.166	0.193

Table 7: TIED COVARIANCE

The result aligns well with our expectations, since a smaller number of components for the fake class and a higher number of components for the authentic one lead to better performance. Going into details we see that the worse results are that of the Full Covariance since it has more parameters to estimate and since it makes strong assumptions about features distributions might not perform well, leading to higher actDCF values. Also, the Tied Covariance assumption of same covariance for each component of a class is not able to capture the feature distributions as well as the others. In conclusion we can say that the best results are obtained by the Diagonal Covariance with his assumption that the feature are conditionally independent given each component and the best choice is 8 components for the fake class and 32 for the authentic one.

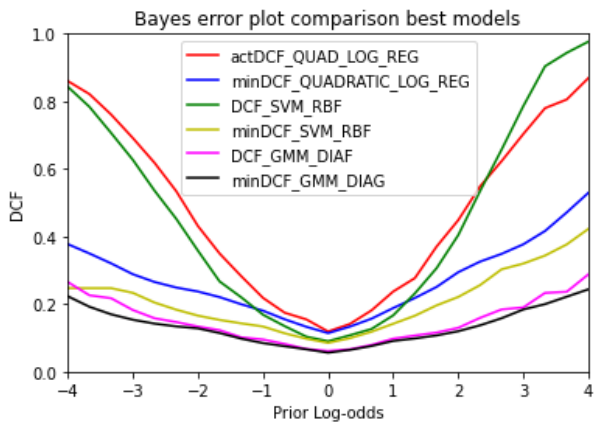
2.7 Final considerations

Considering all the classifiers the most promising method for our application ($\tilde{\pi} = 0.1$) is the last we saw, the Diagonal Covariance.

To complete and perform a more qualitative analysis we report below the Bayes error plot of the best model for each of the main classifier (GMM, logistic regression, SVM) for different applications. Our choices are the Quadratic Logistic Regression, the SVM RBF and the GMM Diagonal:

MODEL	**MIN DCF**
Quadratic Logistic Regression $\lambda \approx 0.031$	0.244
SVM with kernel RBF $K=1, \gamma = e^{-2}$ C= 31,66	0.175
GMM type = Diagonal Covariance components = (8,32)	0.131

We now perform a qualitative analysis visualizing how they behave for different applications:



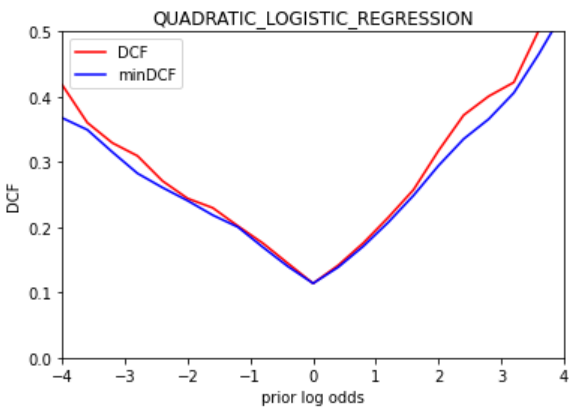
The Bayes plot confirm what is the ranking of the systems suggesting that for the different application the GMM is the model with the best performance and it's also the only one that is well calibrated, while the other two maintains a high actDCf for most of the priors, especially for values outside the range between -1 and 1, so this model could be harmful for applications requiring robust calibration across various operating points.

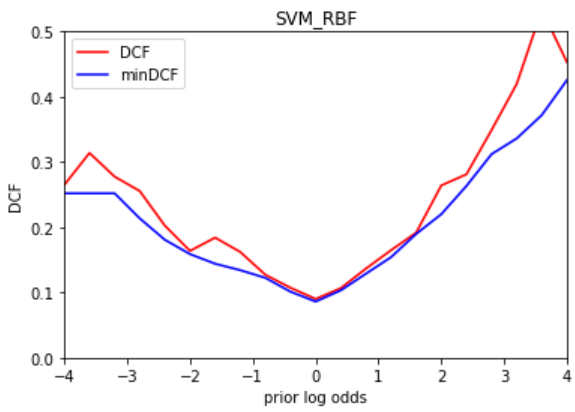
2.8 **Calibration**

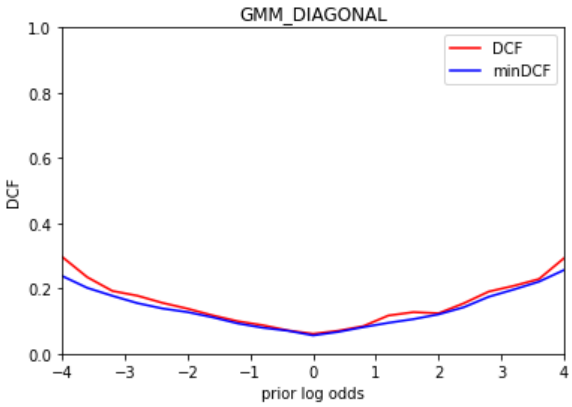
As we have seen, some classifiers may provide scores that are not well-calibrated. In these scenarios we need to either estimate an application-specific threshold or a calibration transformation that maps scores to well-calibrated log-likelihood ratios (LLRs). In the previous chapter to evaluate the different models, we have used only the minimum cost of detection (minDCF), which however depends on a threshold. To understand if the threshold is the theoretical one, we will use a metric called actual DCF (actDCF) which reflects the real performance of the model when we apply it for practical problem. The method that we will adopt to calibrate scores is based on prior-weighted Logistic Regression while to estimate the parameters of the calibration function, we will use a K-Fold approach. We used for the training prior these values: 0.2,0.5,0.9 and then choose for each model the best configuration for the target application. Here we report the best result obtained in terms of actCF and minDCF and what we can observe is that the calibration procedure we applied had a good impact leading to well calibrated scores since the two metrics are close to each other.

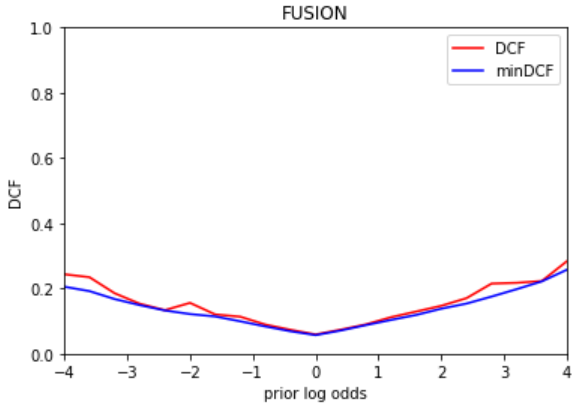
MODEL	**MIN DCF**	**ACT. DCF**
Quadratic Logistic Regression $\lambda \approx 0.031$	0.249	0.272
SVM with kernel RBF $K=1, \gamma = e^{-2}$ C= 31,62	0.169	0.189
GMM type = Diagonal Covariance components = (8,32)	0.132	0.152
Fusion of the three models	0.126	0.157

Below we reported also the plots for the different applications:







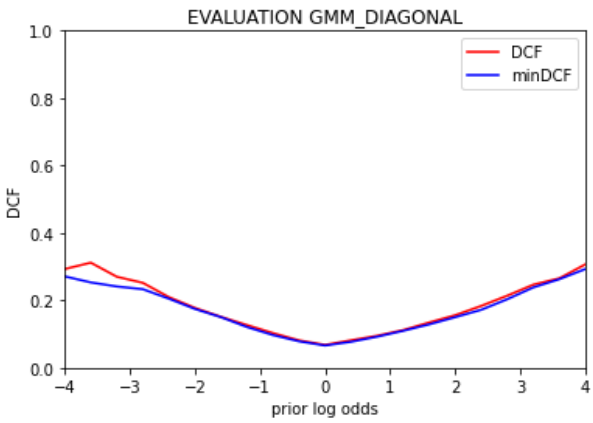


As we can see the Quadratic Logistic Regression and the SVM kernel RBF are now better calibrated for the different applications if we compare the graph above with the one of the previous chapter. The GMM was already well calibrated, despite showing a little miscalibration for values of the log odds bigger than 3, in fact the result obtained remained almost unchanged.

We tried also to evaluate the fusion of the three models but this approach in our case doesn't bring benefits as the actDCF is good but not better than the single systems for the target application so the GMM Diagonal will be the final system that will be evaluated on the application data since it is the one better calibrated and with the lower costs.

Capitolo 3:
EVALUATION

We now analyse performance on the evaluation set to check if the choice and procedures applied during the validation phase were good or we could have done better. We start examining how the delivered model we chose behave with a new dataset: the evaluation dataset, which have the same format of the training dataset we used up to now.

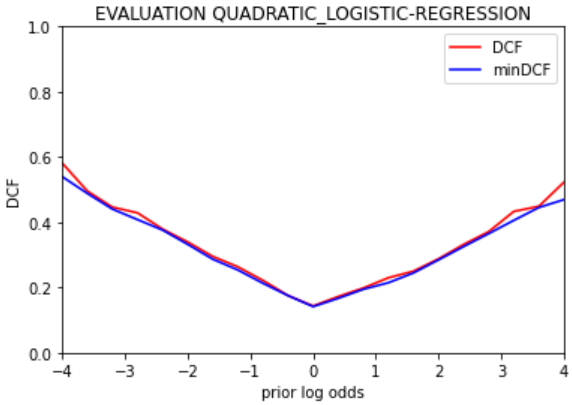


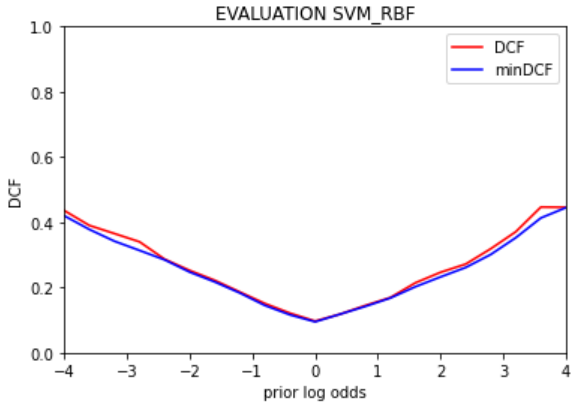
We can see that also for the evaluation dataset the scores provided by the GMM are well calibrated for the different applications. We now compute the minDCF and actDCF of the different systems for the target application in order to assess if our choice was effective.

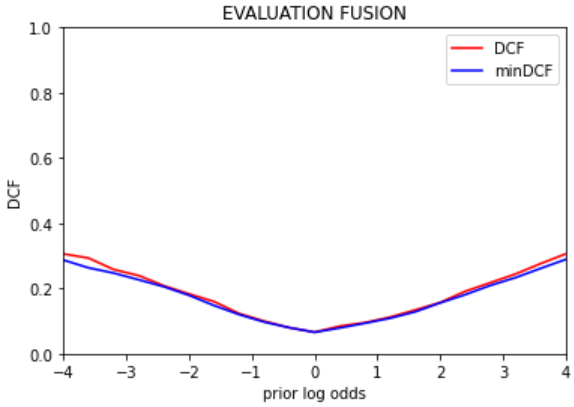
MODEL	**MIN DCF**	**ACT. DCF**
Quadratic Logistic Regression $\lambda \approx 0.031$	0.264	0.368
SVM with kernel RBF $K=1, \gamma = e^{-2}$ C= 31,62	0.264	0.274
GMM type = Diagonal Covariance components = (8,32)	0.188	0.190
Fusion of the three models	0.192	0.204

The GMM with Diagonal Covariance model achieves the lowest actualDCF in both the training (0.152) and evaluation sets (0.190), making it the best performer in terms of actual detection cost. This demonstrates the model's robustness and effectiveness in minimizing the detection cost, which was the key criterion for selection.

We now plot the Bayes Error to well capture if the scores are calibrated for other different applications, and we can see that our strategy was effective.







Our final analysis will consider, for the GMM Diagonal that is our delivered model, all the combinations of components for the two classes we used in the training phase to verify if our choice was effective and optimal also for the evaluation data.

FAKE	AUTHENTIC	VALIDATION MINDCF:	EVALUATION MIN. DCF
4	1	0.145	0.25
4	8	0.140	0.194
4	16	0.137	0.178
8	16	0.132	0.181
8	32	0.131	0.184

Table 7: Evaluation vs Validation

These are some of the most significant results for the two datasets. We can notice that that what we obtained during the Evaluation phase is very similar to the performance we had during the Validation. This suggests that the evaluation population has a distribution similar to that of validation. In both cases the minDCF values showed a decreasing trend as the number of components increased, however while for the validation dataset the lowest minDCF(0.131) was obtained with(8,32) as components the (4,16) performed slightly better on the evaluation set, indicating that this could be a more optimal choice.